



# Application Programming Interface Specifications

---

## Integrated e-filing and CPC 2.0 Project

|                        |                                      |
|------------------------|--------------------------------------|
| <b>API Name</b>        | ITR Life Cycle Status                |
| <b>API Description</b> | API used to get ITR Status lifecycle |

## Version History

| Version | Date       | Description                                                                                                       |
|---------|------------|-------------------------------------------------------------------------------------------------------------------|
| 1.0     | 06/01/2022 | Initial Draft                                                                                                     |
| 1.2     | 3/2//2020  | Exception scenario updated<br>Refund amount remove from activity code text<br>mapping and email Id will be masked |

## Table of Contents

|                                                              |    |
|--------------------------------------------------------------|----|
| 1. Overview.....                                             | 4  |
| 2. About API.....                                            | 4  |
| 3. Target Audience and Pre-requisites .....                  | 4  |
| 4. ITRLifeCycleStatus API Details.....                       | 4  |
| 4.1 API Usage Scenario .....                                 | 5  |
| 4.2 API Request process.....                                 | 5  |
| 4.3 API Protocol.....                                        | 5  |
| 4.4 Request Parameters .....                                 | 5  |
| 4.4.1 Request Header: .....                                  | 5  |
| 4.4.2 Request Body:.....                                     | 6  |
| 4.4.3 Details of data attribute:.....                        | 6  |
| 4.5 Response Parameters.....                                 | 6  |
| 4.6 Activity Code and Text Mapping:.....                     | 9  |
| 4.8 EriITRLifeCycleStatus API - Sample Response format ..... | 11 |
| 5. ITRDownload API Details.....                              | 12 |
| 5.1 API Usage Scenario .....                                 | 12 |
| 5.2 API Request process.....                                 | 12 |
| 5.3 API Protocol.....                                        | 13 |
| 5.4 Request Parameters .....                                 | 13 |
| 5.4.1 Request Header: .....                                  | 13 |
| 5.4.2 Request Body:.....                                     | 13 |
| 5.4.3 Details of data attribute:.....                        | 14 |
| 5.5 Response Parameters.....                                 | 14 |
| 5.6 EriITRDownload API - Sample Request format.....          | 14 |
| 5.7 EriITRDownload API - Sample Response format.....         | 15 |
| 6. API Exception Details.....                                | 15 |

# 1. Overview

ITR Receipt Status APIs is used for displaying already filled ITR Status any time, and all the details as per business requirement during its lifecycle whenever any change will happen in ITR Status.

## 2. About API

|                                                |                                                                                                                                         |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Requester</b>                               | ERI type 2                                                                                                                              |
| <b>Provider</b>                                | ITR Status API                                                                                                                          |
| <b>Description</b>                             | Collection of APIs to enable ERI type-2 to view life cycle of IT returns                                                                |
| <b>Mode of Integration (Real time / Batch)</b> | Real Time                                                                                                                               |
| <b>Processing Details</b>                      | It will give a platform for ERIs to display the lifecycle of ITR filled and display all its statuses during its lifecycle               |
| <b>Pre-Processing Details</b>                  |                                                                                                                                         |
| <b>Service Name</b>                            | EriITRLifeCycleStatus                                                                                                                   |
|                                                | EriITRDownload                                                                                                                          |
| <b>API URL</b>                                 | <a href="http://&lt;hostname&gt;/itrweb/auth/v0.1/client/lifecycle">http://&lt;hostname&gt;/itrweb/auth/v0.1/client/lifecycle</a>       |
|                                                | <a href="http://&lt;hostname&gt;/itrweb/auth/v0.1/returns/downloadItr">http://&lt;hostname&gt;/itrweb/auth/v0.1/returns/downloadItr</a> |

## 3. Target Audience and Pre-requisites

This is technical document and target audiences are ERIs working in their application and interested to integrate their application with IEC 2.0 platform.

The pre-requisites to call this API is that ERI is already registered with IEC 2.0 platform. They have valid credentials to call the API.

## 4. ITRLifeCycleStatus API Details

This service is used to display the lifecycle of ITR filled and display all its statuses during its lifecycle

## 4.1 API Usage Scenario

When ERI application submits the call this API below are the activities are performed:

1. Taxpayer can view/verify the ITR Status of the returns filed through their respective active ERI user.

## 4.2 API Request process

ERI application must have already establish the login session and auth token with eFiling system using the login API. Application will initiate call with API as below:

1. ERI Application will call EriITRLifeCycleStatus API and send the request.

## 4.3 API Protocol

EriITRLifeCycleStatus API is exposed as REST API over the HTTPS. The input data should be sent as JSON document using Content-Type "application/json".

## 4.4 Request Parameters

The request will consist of request header and request body:

### 4.4.1 Request Header:

Header is mandatory and will consists of following values:

**Mandatory Request Header Parameters:**

| Header Name  | Header Value                                                            |
|--------------|-------------------------------------------------------------------------|
| Content-type | application/json                                                        |
| clientId     | clientId value which is provided to ERI as part of the registration     |
| clientSecret | clientSecret value which is provided to ERI as part of the registration |
| authToken    | Auth token from the Login Flow                                          |
| accessMode   | "API"                                                                   |

#### 4.4.2 Request Body:

Request body will consist of below attributes:

1. **data:** data attribute will be Base64 encoded string of API request json. Details of request json attributes are explained in request data element details.
2. **eriUserId:** It is mandatory and valid value is user ID of the ERI
3. **sign:**
  - The API request data attribute should be digitally signed for the message integrity and non-repudiation purposes.
  - Digital signing should always be performed by the ERI from value of data attribute which was generated from request json.
  - The signature should be generated using a valid X.509 certificate
  - signature value should be generated from data field using ERI's DSC private key.
  - ERI should share their DSC public key with ITD to validate the signature.

#### 4.4.3 Details of data attribute:

Below are the request parameters, which is request json used to create data attribute as explained above data attribute of the request body:

| Name of the Parameter | Data type | Max length | Is Mandatory | Description                                                         |
|-----------------------|-----------|------------|--------------|---------------------------------------------------------------------|
| serviceName           | String    | 60         | Yes          | It is mandatory and valid value is " <b>EriITRLifeCycleStatus</b> " |
| pan                   | String    | 10         | Yes          | Valid PAN of the taxpayer.                                          |
| ay                    | String    | 4          | Yes          | 2021                                                                |

#### 4.5 Response Parameters

| Name of the Parameter | Data type | Max length | Is Mandatory | Description |
|-----------------------|-----------|------------|--------------|-------------|
|-----------------------|-----------|------------|--------------|-------------|

|                  |                        |  |     |                                                                                                                            |
|------------------|------------------------|--|-----|----------------------------------------------------------------------------------------------------------------------------|
| messages         | List<String>           |  | Yes | List of information messages. There can be one or more messages. The messages array will be null if there are no messages. |
| errors           | List<Error>            |  | No  | This is a not mandatory field. List of errors. The value will be null if there is no error.                                |
| itrLifeCycleData | List<ItrLifeCycleData> |  | Yes | This is a list of ITR Life Cycle Details for given PAN.                                                                    |
| ay               | String                 |  | Yes | Assessment Year                                                                                                            |
| formTypeCd       | String                 |  | No  | ITR form type code<br>1ITR-1<br>2ITR-2<br>3ITR-3<br>4ITR-4<br>5ITR-5<br>6ITR-6<br>7ITR-7                                   |
| filingTypeCd     | String                 |  | No  | Original/Revised etc<br>O - Original<br>R - Revised                                                                        |
| ackNum           | String                 |  | No  | Acknowledgement Number                                                                                                     |
| ackDt            | Timestamp              |  | No  | Acknowledgement Date                                                                                                       |
| origAckNum       | String                 |  | No  | In case of revised / rectified                                                                                             |
| submitTmstmp     | Timestamp              |  | No  | Submit timestamp                                                                                                           |
| verStatus        | String                 |  | No  | Verification Status                                                                                                        |

|                |                  |    |     |                                                                                                                                                   |
|----------------|------------------|----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------|
| verMode        | String           |    | No  | OTP - Aadhaar OTP<br>NET - Net Banking<br>EVC - EVC<br>BAC - Bank account<br>DAC - Demat account<br>DSC- Digital Signature<br>ITV - ITRV received |
| verDt          | Date             |    | No  | E-verification date                                                                                                                               |
| itrPanDetlList | List<ItrPanDetl> |    | No  | Lists contain the lifecycle of particular ITR                                                                                                     |
| itrActivityCd  | String           |    | No  |                                                                                                                                                   |
| activityDt     | Time Stamp       |    | No  |                                                                                                                                                   |
| activityTxt    | String           |    | No  | Will contain optionally relevant fields as applicable in JSON format.<br>(refer mapping - Activity Code and Text Mapping)                         |
| statusDesc     | String           |    | No  |                                                                                                                                                   |
| successFlag    | Boolean          |    | Yes | Indicates whether EriITRLifeCycleStatus call is successful. Possible values is true or false.<br><b>"successFlag": true</b>                       |
| httpStatus     | String           | 20 | No  | http codes corresponding to response. Possible value is "OK"                                                                                      |

## 4.6 Activity Code and Text Mapping:

| itr_activity_cd | activityTxt                                                                   |
|-----------------|-------------------------------------------------------------------------------|
| 1               | itrvRecieptDt, commRefNo, emailDt, emailId, smsDt                             |
| 2               | itrvRecieptDt, commRefNo, emailDt, emailId, smsDt                             |
| 3               | evcAcceptDt                                                                   |
| 4               | itrvAcceptDt                                                                  |
| 8               | rejectDt, reasonCode, commRefNo, emailDt, emailId, smsDt, reasonCode          |
| 21              | defectiveCommDt, emailDt, emailId                                             |
| 22              | defectiveCommDt, emailDt, emailId                                             |
| 23              | defectiveCommDt, emailDt, emailId                                             |
| 31              | aoTransDate, aoCode, aoDesc                                                   |
| 44              | commRefNo, emailDt, emailId                                                   |
| 50              | commRefNo, emailDt, emailId, intimationDt                                     |
| 61              | orderDt, smsDt, emailDt, emailId, speedPostDetl                               |
| 62              | orderDt, smsDt, emailDt, emailId                                              |
| 63              | orderDt, smsDt, emailDt, emailId                                              |
| 64              | orderDt, smsDt, emailDt, emailId                                              |
| 65              | orderDt, smsDt, emailDt, emailId, paymentMode                                 |
| 71              | orderDt, smsDt, emailDt, emailId, speedPostDetl                               |
| 72              | orderDt, smsDt, emailDt, emailId                                              |
| 73              | orderDt, smsDt, emailDt, emailId                                              |
| 74              | orderDt, smsDt, emailDt, emailId                                              |
| 75              | orderDt, smsDt, emailDt, emailId, paymentMode                                 |
| 100             | rectDt                                                                        |
| 108             | rejectDt, rejectReason                                                        |
| 131             | aoTransDate, aoCode, aoDesc                                                   |
| 211             | invalidationDt, invaliditingReason, communDt, emailDt, emailId, speedPostDetl |
| 214             | commRefNo, emailDt, emailId                                                   |

|     |                                               |
|-----|-----------------------------------------------|
| 600 | refundReIssue                                 |
| 613 | orderDt, smsDt, emailDt, emailId, paymentMode |
| 619 | speedPostNo, deliveryDt, aoCode, aoDesc       |
| 620 | dispatchDt                                    |
| 621 | paymentMode, clearanceDt                      |
| 622 | failureDt, reasonCode, communDt, emailId      |
| 623 | failureDt, reasonCode, communDt, emailId      |
| 624 | failureDt, reasonCode, communDt, emailId      |
| 625 | failureDt, reasonCode, communDt, emailId      |
| 626 | reasonCode, rejectDt, commRefNo               |
| 627 | L1_L2_communDt, reissueReqDt, status, reason  |
| 628 | paymentMode, clearanceDt                      |
| 713 | revertDt, reason                              |
| 714 | revertDt, reason                              |
| 731 | aoTransDate, aoCode, aoDesc                   |
| 984 | failureDt, reasonCode, communDt, emailId      |
| 985 | failureDt, reasonCode, communDt, emailId      |
| 989 | failureDt, reasonCode, communDt, emailId      |
| 990 | failureDt, reasonCode, communDt, emailId      |
| 997 | eVrfDt, modeEVrf                              |

#### 4.7 EriITRLifeCycleStatus API - Sample Request format

```
{
  "data": "",
  "sign": "",
  "eriUserId": ""
```

```
}
```

**data** tag will be Base64Encoded string from following request json

```
{  
  "serviceName" = "EriITRLifeCycleStatus",  
  "pan" = null,  
  "ay" = "2021"  
}
```

## 4.8 EriITRLifeCycleStatus API - Sample Response format

```
{  
  "messages": null,  
  "errors": null,  
  "itrLifeCycleData": [  
    {  
      "ay": "",  
      "formTypeCd": "",  
      "filingTypeCd": "R",  
      "ackNum": "",  
      "ackDt": "2021-11-15",  
      "submitTmstp": "2021-11-15 12:26:22.908",  
      "verStatus": "Y",  
      "verDt": "2021-11-16",  
      "itrPanDetlList": [  
        {  
          "itrActivityCd": "997",  
          "activityDt": "2021-11-16",  
          "activityTxt": {},  
          "statusDesc": "Successfully e-verified"  
        },  
        {  
          "itrActivityCd": "998",  
          "activityDt": "2021-11-15",  
          "activityTxt": {},  
          "statusDesc": "Pending for e-verification"  
        },  
        {  
          "itrActivityCd": "999",  
          "activityDt": "2021-11-15",  
          "activityTxt": {},  
          "statusDesc": "ITR Filed"  
        }  
      ]  
    },  
    {  
      "ay": "",  
      "formTypeCd": "",  
      "filingTypeCd": "O",
```

```

    "ackNum": "",
    "ackDt": "2021-11-15",
    "submitTmstp": "2021-11-15 12:11:19.062",
    "verStatus": "Y",
    "verDt": "2021-11-16",
    "itrPanDetlList": [
      {
        "itrActivityCd": "997",
        "activityDt": "2021-11-16",
        "activityTxt": {},
        "statusDesc": "Successfully e-verified"
      },
      {
        "itrActivityCd": "998",
        "activityDt": "2021-11-15",
        "activityTxt": {},
        "statusDesc": "Pending for e-verification"
      },
      {
        "itrActivityCd": "999",
        "activityDt": "2021-11-15",
        "activityTxt": {},
        "statusDesc": "ITR Filed"
      }
    ]
  },
  "successFlag": true,
  "httpStatus": "ACCEPTED"
}

```

## 5. ITRDownload API Details

This service is used to download ITR filled.

### 5.1 API Usage Scenario

When ERI application submits the call this API below are the activities are performed:

1. Taxpayer can download the ITR any time

### 5.2 API Request process

ERI application must have already establish the login session and auth token with eFiling system using the login API. Application will initiate call with API as below:

1. ERI Application will call EriITRDownload API and send the request.

## 5.3 API Protocol

EriITRDownload API is exposed as REST API over the HTTPS. The input data should be sent as JSON document using Content-Type "application/json".

## 5.4 Request Parameters

The request will consist of request header and request body:

### 5.4.1 Request Header:

Header is mandatory and will consists of following values:

#### Mandatory Request Header Parameters:

| Header Name  | Header Value                                                            |
|--------------|-------------------------------------------------------------------------|
| Content-type | application/json                                                        |
| clientId     | clientId value which is provided to ERI as part of the registration     |
| clientSecret | clientSecret value which is provided to ERI as part of the registration |
| authToken    | Auth token from the Login Flow                                          |
| accessMode   | "API"                                                                   |

### 5.4.2 Request Body:

Request body will consist of below attributes:

1. **data:** data attribute will be Base64 encoded string of API request json. Details of request json attributes are explained in request data element details.
2. **eriUserId:** It is mandatory and valid value is user ID of the ERI
3. **sign:**
  - The API request data attribute should be digitally signed for the message integrity and non-repudiation purposes.
  - Digital signing should always be performed by the ERI from value of data attribute which was generated from request json.
  - The signature should be generated using a valid X.509 certificate
  - signature value should be generated from data field using ERI's DSC private key.
  - ERI should share their DSC public key with ITD to validate the signature.

### 5.4.3 Details of data attribute:

Below are the request parameters, which is request json used to create data attribute as explained above data attribute of the request body:

| Name of the Parameter | Data type | Max length | Is Mandatory | Description                                                        |
|-----------------------|-----------|------------|--------------|--------------------------------------------------------------------|
| serviceName           | String    | 60         | No           | It is not a mandatory and valid value is " <b>EriITRDownload</b> " |
| pan                   | String    | 10         | Yes          | Valid PAN of the taxpayer.                                         |
| ackNum                | String    |            | Yes          | Acknowledgement Number                                             |
| format                | String    | 4          | Yes          | PDF or JSON                                                        |

## 5.5 Response Parameters

| Name of the Parameter | Data type    | Max length | Is Mandatory     | Description                                                                                                                |
|-----------------------|--------------|------------|------------------|----------------------------------------------------------------------------------------------------------------------------|
| messages              | List<String> |            | Yes (only Error) | List of information messages. There can be one or more messages. The messages array will be null if there are no messages. |
| errors                | List<Error>  |            | No (only Error)  | This is a not mandatory field. List of errors. The value will be null if there is no error.                                |
| successFlag           | Boolean      |            | Yes (only Error) | Indicates whether EriITRDownload call is successful. Possible values is false. <b>"successFlag": false</b>                 |
| httpStatus            | String       | 20         | No (only Error)  | http codes corresponding to response. Possible values is "OK"                                                              |
| file                  | Byte Array   |            | Yes (on Success) | Binary Format PDF / Binary Format JSON                                                                                     |

## 5.6 EriITRDownload API - Sample Request format

{ "data": "", "sign": "", "eriUserId": "" } **data** tag will be Base64Encoded string from following request json{  
"serviceName" = "EriITRDownload",  
"pan" = null,  
"ackNum" = null,  
"format" = "PDF" }

## 5.7 EriITRDownload API - Sample Response format

On Error:

```
{  
  "messages": [  
    {  
      "code": "EF500058",  
      "type": "ERROR",  
      "desc": "The PAN is not a valid client for this ERI.",  
      "fieldName": null  
    }  
  ],  
  "errors": [],  
  "successFlag": false,  
  "header": {  
    "formName": null  
  }  
}
```

On Success:

<<Downloadable Password Protected PDF in binary format. Password will be PAN continued by DOB/DOI in DDMMYYYYY format>>

Json Response:

```
{ "ITR": "<<JSON encrypted with symmetric key>>" }
```

## 6. API Exception Details

| Scenario                                        | Error code     | Error String       | Message Type |
|-------------------------------------------------|----------------|--------------------|--------------|
| When any attributes are missing in request JSON | <i>EF40000</i> | JSON data invalid. | ERROR        |

|                                                                       |          |                                                                                                |       |
|-----------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------|-------|
| When attributes are incorrect in JSON request data                    | EF20123  | Invalid Request data.                                                                          | ERROR |
| When PAN is not in a valid format                                     | EF00011  | Please enter a valid PAN Number.                                                               | ERROR |
| When the user is inactive.                                            | EF00098  | The PAN entered is inactive. Please contact your Accessing Officer to activate the PAN.        | ERROR |
| When return is not filed through the ERI for given PAN                | EF500099 | No return is filed by the ERI for this PAN for given AY                                        | ERROR |
| When AY is before 2021                                                | EF500098 | No returns are filed by ERI for this AY                                                        | ERROR |
| When no records are found for given PAN and ERI                       | EF500100 | No returns are filed for this PAN for given AY                                                 | ERROR |
| No ITR Receipt Status found                                           | EF30018  | No Result found                                                                                | INFO  |
| When ERI client linkage fails                                         | EF500058 | The PAN is not a valid client for this ERI.                                                    | ERROR |
| When user enters incorrect arnNumber                                  | EF20038  | Invalid Receipt number. Please retry and enter correct acknowledgement number                  | ERROR |
| When arnNumber is not accessible for Eri user                         | EF500053 | Acknowledgement form cannot be generated for this ITR as it is not submitted through this ERI. | ERROR |
| When Acknowledgement number is not for the entered PAN                | EF500101 | The entered Acknowledgement number is not linked to the given PAN, please enter correct PAN    | ERROR |
| When PAN is null                                                      | EF20008  | PAN cannot be null or empty .Please enter correct PAN.                                         | ERROR |
| When Acknowledgement number is null                                   | EF10007  | Acknowledgement Number cannot be null or empty                                                 | ERROR |
| When format is null or when value of format is other than pdf or JSON | EF500089 | Please provide valid value for format.                                                         | ERROR |
| When ERI client linkage fails                                         | EF500058 | The PAN is not a valid client for this ERI.                                                    | ERROR |
| When any server-side failure occurs                                   | EF500062 | Request not completed due to technical issue. Please try again.                                | ERROR |